ENDPOINT SECURITY POSTURE MONITORING USING SPLUNK

Project Type: Security Operations Center (SOC) – Beginner Level
Tools Used: Splunk Enterprise, Search Processing Language (SPL), CSV Dataset
Project Focus: Endpoint Security Monitoring and Risk Assessment

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1. Introduction

Endpoint security is an important part of cybersecurity because computers and other devices can become targets of malware, unauthorised access and other security threats.

This project focuses on analysing endpoint security information using Splunk Enterprise. A CSV dataset containing endpoint details, health status, antivirus protection, encryption status and Tamper Protection information was imported into Splunk.

Search Processing Language (SPL) was used to investigate the data, identify potential security weaknesses and create a dashboard to visualise endpoint security information.

2. Project Objectives

The main objectives of this project were:

- Understand how Splunk is used in security monitoring.
- Import endpoint security data into Splunk.
- Write SPL queries to investigate endpoint information.
- Monitor endpoint health status.
- Identify devices with Tamper Protection disabled.
- Analyse available encryption information.
- Review antivirus protection distribution.
- Identify endpoints requiring further investigation.
- Create an endpoint security monitoring dashboard.

3. Tools and Technologies Used

Tool	Purpose
Splunk Enterprise	Security data analysis and monitoring
SPL	Searching, filtering and analysing endpoint data
CSV Dataset	Endpoint security information
GitHub	Project documentation and portfolio hosting

4. Dataset Description

The project used a CSV dataset containing seven endpoint records.

The dataset included information such as:

- Device Name
- Operating System
- Health Status
- Antivirus Protection
- Encryption
- Last Active
- Tamper Protection

The dataset contained Windows and macOS endpoint records.

5. Data Ingestion into Splunk

The CSV dataset was uploaded into Splunk Enterprise using the data upload functionality.

The following configuration was used:

- Index Name: av
- Source Type: csv
- Data Format: CSV

After uploading, the imported records were verified using an SPL search.

index=av sourcetype=csv

The search successfully returned seven endpoint records.

6. Investigation and SPL Queries

6.1 Endpoint Health Status Analysis

index=av sourcetype=csv

| stats count by "Health Status"

Purpose:
To identify the number of endpoints classified under each health status.

Finding:

- 6 endpoints were marked Healthy.
- 1 endpoint was marked Critical.

6.2 Tamper Protection Analysis

index=av sourcetype=csv

| stats count by "Tamper protection"

Purpose:
To identify endpoints with Tamper Protection enabled or disabled.

Finding:

- 4 endpoints had Tamper Protection On.
- 3 endpoints had Tamper Protection Off.

6.3 Endpoint Security Inventory

index=av sourcetype=csv

| table Name OS "Health Status" Protection Encryption "Tamper protection" "Last active"

Purpose:
To display endpoint security information in a structured table.

6.4 Encryption Status Analysis

index=av sourcetype=csv

| eval Encryption_Status=if(isnull(Encryption) OR Encryption="", "Not Reported", Encryption)

| stats count by Encryption_Status

Purpose:
To examine the availability of encryption information.

Finding:

- 1 endpoint had encryption marked Enabled.
- 6 endpoints had encryption information not reported.

6.5 Antivirus Protection Distribution

index=av sourcetype=csv

| stats count by Protection

Purpose:
To examine the distribution of endpoint protection products.

Finding:

- 6 endpoints used Intercept X Advanced with XDR.
- 1 endpoint used Intercept X Advanced.

6.6 Endpoint Last Active Analysis

index=av sourcetype=csv

| table Name "Health Status" "Last active"

| sort "Last active"

Purpose:
To review the recorded last-active dates of endpoints.

Finding:
The dataset contained last-active dates ranging from September 28, 2025, to October 4, 2025.

6.7 Combined Endpoint Risk Assessment

index=av sourcetype=csv

| eval Tamper_Status=if('Tamper protection'="Off","Disabled","Enabled")

| eval Encryption_Status=if(isnull(Encryption) OR Encryption="","Not Reported",Encryption)

| eval Risk_Flag=if('Health Status'="Critical" OR Tamper_Status="Disabled" OR Encryption_Status="Not Reported","Review Required","No Listed Issue")

| table Name OS "Health Status" Tamper_Status Encryption_Status Risk_Flag

Purpose:
To combine endpoint health, Tamper Protection and encryption information to identify devices requiring further review.

Finding:

- 6 endpoints were flagged as Review Required.
- 1 endpoint was classified as No Listed Issue.

6.8 Overall Endpoint Risk Summary

index=av sourcetype=csv

| eval Tamper_Status=case('Tamper protection'="Off","Disabled",'Tamper protection'="On","Enabled",true(),"Not Reported")

| eval Encryption_Status=if(isnull(Encryption) OR Encryption="","Not Reported",Encryption)

| eval Risk_Flag=if('Health Status'="Critical" OR Tamper_Status="Disabled" OR Encryption_Status="Not Reported","Review Required","No Listed Issue")

| stats count by Risk_Flag

Purpose:
To calculate the number of endpoints in each risk category.

Finding:

- Review Required: 6 endpoints.
- No Listed Issue: 1 endpoint.

7. Dashboard Development

An Endpoint Security Monitoring dashboard was created in Splunk to present the investigation results.

The dashboard contains the following visualisations:

1. Endpoint Health Status
2. Tamper Protection Status
3. Endpoint Security Inventory
4. Endpoint Encryption Status
5. Antivirus Protection Distribution
6. Endpoint Last Active Overview
7. Combined Endpoint Risk Assessment
8. Overall Endpoint Risk Summary

The dashboard provides a consolidated view of the endpoint security information and helps identify devices requiring attention.

8. Key Security Findings

The investigation identified the following observations:

Security Metric	Result
Total Endpoints	7
Healthy Endpoints	6
Critical Endpoints	1
Tamper Protection Disabled	3
Tamper Protection Enabled	4
Encryption Enabled	1
Encryption Not Reported	6
Review Required	6
No Listed Issue	1

Important Observations

1. Critical Endpoint

One endpoint was classified as Critical. It should be prioritised for further investigation.

2. Tamper Protection Disabled

Three endpoints had Tamper Protection disabled. This configuration should be reviewed because it may reduce protection against unauthorised changes to security software.

3. Missing Encryption Information

Six endpoints did not have reported encryption status. Further verification is required to determine whether disk encryption is enabled.

4. Overall Endpoint Risk

Six endpoints were flagged for review based on the defined investigation rules.

These findings represent configuration and health observations from the supplied dataset. They do not independently establish that an endpoint was compromised.

9. Recommendations

Based on the investigation, the following recommendations are suggested:

1. Investigate the endpoint marked Critical.
2. Review and enable Tamper Protection on devices where it is disabled, following organisational procedures.
3. Verify disk encryption status on endpoints where information is missing.
4. Regularly monitor antivirus protection status.
5. Review endpoint health information periodically.
6. Investigate outdated last-active records using current endpoint telemetry.
7. Maintain regular endpoint security monitoring through Splunk dashboards.

10. Limitations

- The project uses a sample dataset containing only seven endpoint records.
- The information represents the dataset's recorded values and may not reflect current endpoint conditions.
- Missing encryption information cannot be interpreted as encryption being disabled.
- The risk classification is based on predefined rules and is not a confirmed security incident.
- No live endpoint remediation or malware investigation was performed.

11. Learning Outcomes

Through this project, I gained practical experience in:

- Using Splunk Enterprise.
- Importing and verifying CSV data.

- Understanding and writing SPL queries.
- Analysing endpoint security configurations.
- Identifying potential security weaknesses.
- Performing basic endpoint risk assessment.
- Creating security monitoring dashboards.
- Documenting security investigation findings.

12. Conclusion

This project provided practical exposure to basic SOC Analyst activities using Splunk Enterprise.

By importing endpoint security data, writing SPL queries and creating a monitoring dashboard, I was able to analyse endpoint health, Tamper Protection, encryption information and antivirus protection.

The investigation identified six endpoints requiring further review and one endpoint with no listed issue under the checks performed.

Overall, this project helped develop foundational skills in security monitoring, data analysis, endpoint security assessment and SOC investigation documentation.

Project Status: Completed – Beginner-Level SOC Portfolio Project

Prepared for: Cybersecurity / SOC Analyst Portfolio